URL-based Phishing Websites Detection via Machine Learning

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***Abstract*—Phishing is a cybersecurity attack that is used to trick victim users to provide sensitive information or deploy malicious software on their infrastructure. Depending on the target system and users, these attacks can inflict severe negative impacts on the system. Therefore, researchers have been working on developing phishing detection and prevention techniques to thwart these attacks. In this paper, we present an efficient phishing websites detection system that analyzes the phishing websites URL addresses to learn data patterns that can identify authentic and phishing websites. Our system employs machine learning techniques such as neural networks and decision trees to learn data patterns in websites URLs. We evaluate our system on a recent phishing websites dataset using classification accuracy as a performance indicator. Our best result shows that decision trees models provide 97.40% classification accuracy on the almost balanced-class dataset.**

Keywords— Cybersecurity, Phishing, Shallow Neural Network, Decision Tree, Classification, Accuracy

I. Introduction

Phishing is a social engineering cybersecurity attack that involves an attacker who provides a counterfeit piece of information that is hand-crafted skillfully to trick a user (human victim usually) to provide sensitive information to the attacker or to install malicious software on the victim's computing platform [1]. The simplest scenario of a phishing attack is shown in Fig. 1. The attacker first creates an exact copy of the target website (bank, email server, government website … etc.) that is fully under his control. Next, he sends a carefully spoofed email making it look as though it is sent by an authentic organization such as governmental bodies, healthcare institutions, banks, or credit card companies. The email might ask the user to provide personal information or it might contain a link to the fake website, asking the victim user to log in to the targeted website. If the victim user accesses the target website via the provided link, she will be directed to the fake website which is controlled by the attacker. The victim's log-in credentials are collected by the attacker who can use them to access the authentic website. The literature includes several types of phishing attacks such as email phishing [2], Short Message Service (SMS) phishing [3], voice phishing (also known as vishing) [4], and web page hijacking [5].

While at first sight, phishing might be perceived as a low risk attack since it usually targets a small number of users, it can have very severe consequences. For instance, a phishing attacker succeeded to steal 100 million USD from Google and Facebook using fake invoices [6]. Another more recent phishing attack spear-phished the CEO of Crelan bank in Belgium to steal 75.8 million USD. What is even more striking about this incident is that it was launched from a foreign country and the attacker identity is still unknown. This indicates that such attacks can have severe negative impacts on individuals and organizations which necessities further cybersecurity efforts to thwart such attacks. Therefore, it is of crucial importance to develop detection techniques that can recognize phishing websites and alert the user or system administrator before the attack is launched.

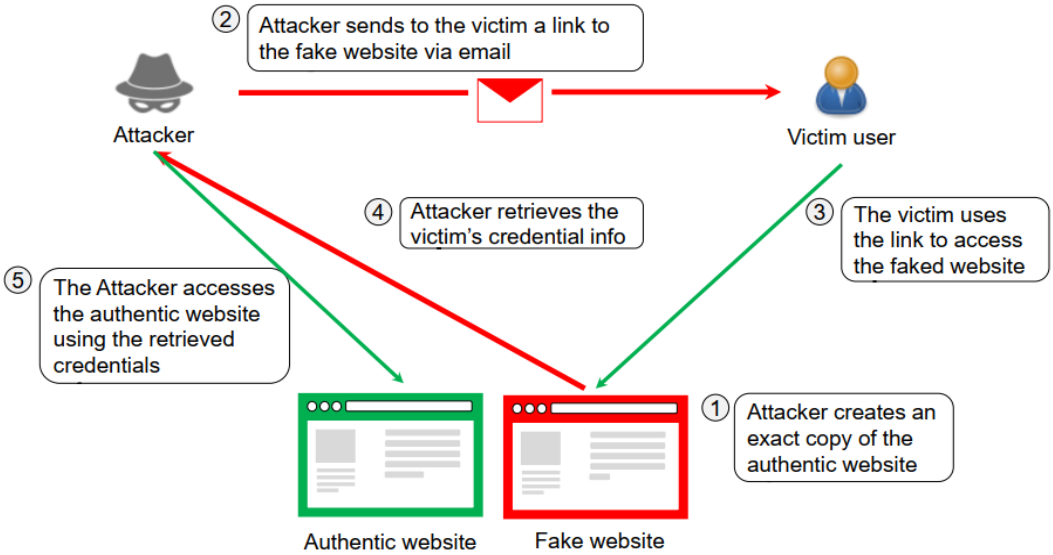

Fig.1. Phishing attack via a fake website.

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In this work, we aim at providing an efficient detection technique for identifying phishing websites by analyzing the phishing websites URLs. We use machine learning as a framework for learning data patterns hidden in phishing websites URLs. More specifically, we use Shallow Neural Networks (SNNs) and Decision Trees (DT) in our detection system as they proved to be fast to train and test and provide high classification accuracy results. We employ the Phishing Websites Dataset (PWS), a newly compiled dataset that contains a collection of authentic and phishing websites samples proposed in 2020 [8]. The dataset is categorized into a small dataset (balanced-class) and a large or full (unbalanced-class) dataset. Detailed description of the dataset is provided in Section 2. Our system is evaluated on both datasets using classification accuracy as a performance indicator. The DT models have been found to outperform the SNN models scoring 97.4% and 96.3% on the small and large datasets, respectively. The best results of different SNN models scored 92.8% 89.7% on the small and large datasets, respectively. It should be noted that, to the best of our knowledge, our work is the first work that employs this dataset for the phishing detection problem. This gives our work more gravity due to the importance of evaluating this newly compiled dataset.

A. Our contributions.

In brief, our contributions can be summarized as follows:

a) We provide a machine learning-based detection system of phishing websites using URL addresses.
b) We use SSNs and DTs supervised learning algorithms to classify the websites into authentic and phishing websites.
c) We evaluate our system on the two categories of phishing websites dataset. To the best of our knowledge, this is the first evaluation of these new datasets.
d) Our analysis shows that DTs provide better performance on both datasets scoring 95.5% and 93.7% classification accuracy on the small and large datasets, respectively.

B. Organization of the paper.

The paper is organized as follows. Section 2 introduces the dataset used in this work. We provide the system model, testbed environment and configurations in Section 3. In Section 4, we present our results and analyze the performance of the proposed system. Finally, Section 5 concludes the work.

II. Dataset of Phishing Activity

The phishing websites dataset [8] is used to evaluate the performance of our system. This dataset comes in two variations; small and large datasets. The authors collected the websites information from two sources; a) Phishtank [9] an organization that keeps track of phishing websites that adopts a community-based phishing verification system where users enquire on suspected phishing websites and other community users verify and suggest whether it is a phish or not, and b) Alexa [10] an organization for websites ranking that keeps track of authentic websites. The authors studied 111 features in websites URL addresses. 96 features are extracted directly from the website ULR address, whereas the rest (15 features) are derived using a custom Python code.

The authors extracted features from the whole URL address such as number of specific special characters (like '.', '$', '#' … etc.), and from sub-regions of the URL address such as the domain URL, the URL directory, and the URL file name. They also extracted features that are specific to each region of the URL such as the domain length and IP format, directory and file name lengths. The regions of the URL address are depicted in Figure 2. The remaining 15 features includes attributes related the user experience and behavior of the website such as response time, number of redirects, URL shortening and so on. The reader is referred to the referenced dataset for more details on dataset compilation process and the feature extracted.

The full specifications of the datasets are provided in Table 1. It can be seen that the small dataset is almost a balanced dataset. The most noticeable feature of the datasets is that the small dataset is almost balanced, i.e., number of samples in each class are almost equal, whereas the large dataset is unbalanced and more biased towards the authentic class.

TABLE I. PHISHING WEBSITES DATASET STATISTICS.

Specification	Small dataset	Large dataset
No. of samples	58,645	88,647
No. authentic websites	27,998	58,000
No. phishing websites	30,647	30,647
No. of features	111	111

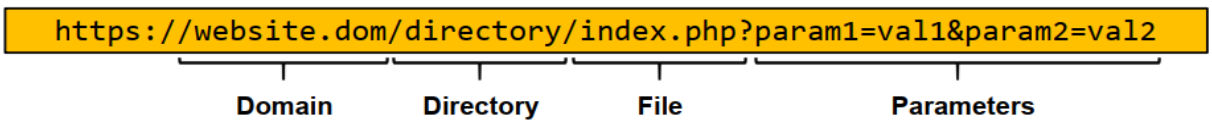

Fig. 2. URL address sub-regions (The figure inspired by Fig. 1 in [8]).

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III. System Modeling and Environment

Machine learning techniques are usually employed by establishing an explicit model aiming to analyze and recognize the data patterns in a supervised or unsupervised fashion [11]. Phishing websites detection can be considered as a typical classification problem that belongs to supervised machine learning model [12]. Hence, a labeled dataset is required for model training, validation and testing. In this paper, we are specifically interested in experimentally investigating and developing an automated intelligent detection system that analyzes the phishing websites URL addresses to learn data patterns that can identify authentic and phishing websites for improved communication processes. A number of machine learning approaches have been used to provide detect phishing websites based on prescribed datasets [13-17]. In general, these models are composed of three general modules as shown in Fig. 3 below and explained as follows:

- Preprocessing: data records are acquired and collected from internet environment, then structured within an organized dataset. Also, feature selection (selecting the most appropriate variables for the task at hand), data conversion, encoding, normalization, shuffling, augmentation and distribution (breaking data into training, validation and testing datasets) are all performed at this module [18].
- Training: a machine learning algorithm is adopted to characterize the patterns of various types of data, and build a corresponding system model. Usually, 70% of the dataset are used as a training data to train the model, and 30 % of the dataset are used as testing/validation data to test the accuracy of the model's predictions. The learning step involves model experimentation, training, building, and testing.
- Detection: The application step which involves model deployment and prediction. Once the model is constructed, the observed website URLs will be used as model input to recognize its pattern and compare it to the patterns generated by the trained/validated model. If the pattern of the observation is recognized to be matched with an existing phishing URLs, risk notification will be triggered to inform about the discovered phishing website.

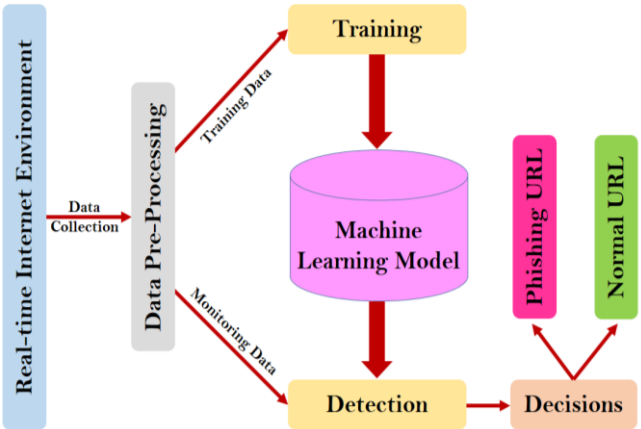

Fig. 3. Architecture of Machine learning based Phishing websites detection

More specifically, Fig.4 demonstrates the comprehensive workflow diagram to develop the proposed detection system for the phishing websites. The proposed detection workflow started by adopting the aforementioned phishing websites dataset to be learned by the intelligent detection module. However, the instances of the dataset are undergone a data preprocessing operations which are essential phases for the detection task involved in this work.

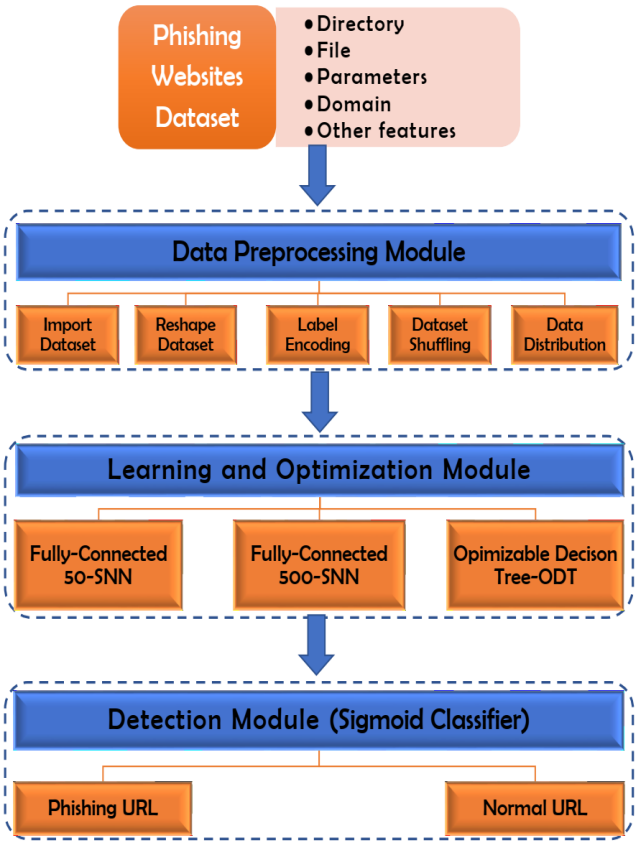

Fig. 4. Workflow Diagram to Develop the Proposed Classification Model.

According to the workflow diagram, the preprocessing module is mainly comprised of five successive processes that are applied on the original records of the phishing websites dataset in order to articulate these records in an acceptable numerical features that can be fed and handled by machine learning techniques located at the next modules to provide proper inferences and decisions. For the next stage, the learning and optimization module, we have examined the preprocessed dataset using three different machine learning models including Narrow Shallow Neural Network (N-SNN), Wide Shallow Neural Network (W-SNN), and the Optimizable Decision Tree/ GentleBoost Ensemble (ODT) [19, 20]. For the last stage, the detection module, we have employed the sigmoid classifier to produce the final output decisions as either phishing URL or normal URL. Besides, Table 2 summarizes the system modeling and environment configurations and parameters.

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TABLE II. SUMMARY OF SYSTEM MODELING AND ENVIRONMENT CONFIGURATIONS AND PARAMETERS.

Specifications	Description
Dataset Importation	Dataset imported as .CSV type to MATLAB tables then converted to double matrices.
Dataset Statistics	Small Dataset (58,645 Records) Large Dataset (88,647 Records)
Target Encoding	One Hot Encoding for the two classes of target as follows: Phishing URL (01), Normal URL (10)
Learning Models	N-SNN with 50-Hidden Neurons, One-Neuron Output Layer. W-SNN with 500-Hidden Neurons, One-Neuron Output Layer. ODT with 923 -3758 of Splits using Gini-Diversity Index.
Dataset Distribution	Randomly divide targets into 3-sets as follows: Training: 80%, Validating: 5%, Testing: 15%
Optimization Techniques	Scaled Conjugate Gradient Backpropagation (SCGB) [21] For SNN, Bayesian Gradient Optimization (BGO) [22] for ODT
Performance Analysis	Cross-Entropy Loss (LCE) Function [23] for SNN. Mean Squared Error (MSE) Function [24] for ODT.
Validation Policy	6-Validation Checks and 6-Fold Cross Validation [25]. Validations are randomly executed at every run.
Max Number of Epochs	170 Epochs for N-SNN/ Shuffling Performed at Every Epoch 218 Epochs for W-SNN/ Shuffling Performed at Every Epoch 30 Epochs for ODT/ Shuffling Performed at Every Epoch

IV. RESULTS AND DISCUSSION

Evaluating the performance of detection system is an essential task of machine learning models. Large number of evaluation metrics are available to measure effectiveness of the machine learning models. As such, the confusion Matrix and its associated elements (i.e., True positives, True negatives, False positives, False negatives), detection error, and detection accuracy [26]. Such results are reported in Table 3, summery of evaluation metrics. As can be clearly inferred from the table, our best performance results have been recorded for the decision trees models scoring 97.40% classification accuracy on the almost balanced-class dataset. Besides, our simulation results reported an average prediction speed (A.V.S) as a measure of the number of observations processed per second to be in the range (170,000 obs/sec - 240,000 obs/sec). This means that the average prediction overhead (A.P.O) can be calculated as follows:

$$A.P.O = 1 / (A.V.S) = 1 / (205{,}000) = 4.88\ \mu.sec$$

TABLE III. SUMMARY OF EVOLUTION METRICS' RESULTS FOR THE THREE ML-MODELS.

Dataset	Model	Depth	Error (%)	Accuracy (%)	F1 Score (%)
Large	N-SNN	50 Neurons	7.4	92.6	89.6
	W-SNN	500 Neurons	7.2	92.8	89.9
	O-DT	923 Splits	4.5	97.4	96.3
Small	N-SNN	50 Neurons	10.8	89.2	89.9
	W-SNN	500 Neurons	10.3	89.7	90.4
	O-DT	3758 Splits	6.3	94.9	95.2

In figure 5, we demonstrate the general arrangement of the two-class confusion matrix that can be used for detection models providing four types of detection results with different color-conventions, including true positive (green), true negative (green), false positive (red), and false negative (red). The first two (green cells) represents correct classification, while the last two (red cells) represent classification errors.

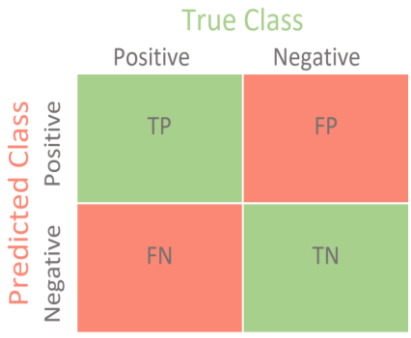

Fig. 5. General Form of Two-Class Confusion Matrix [27]

Subsequently, we provide the overall results of the two-class confusion matrices attained for our phishing detection models using N-SNN, W-SNN and O-DT in Figure 6. Based on the numerical outcomes of these matrices, we can obviously observe the high performance of our detection models in predicting the phishing URLs using the identified phishing websites datasets as most of the data instances are correctly classified (See green areas).

Moreover, the receiver operating characteristic (ROC) curves [28] of our two-classes classifier for the overall datasets are illustrated in Fig. 6. ROC curve plots parametrically TPR(T) versus FPR(T) with T as the varying parameter. Since almost experiments yield a point in the upper left corner (0,1) of the ROC space, the classifier is approaching the excellent classification case recording 91%, 92%, and 96% of the area under the curve (AUC) of the small dataset system for the N-SNN model, W-SNN model, and O-DT model, respectively, and recording 93%, 94%, and 97% of the area under the curve (AUC) of the large dataset system for the N-SNN model, W-SNN model, and O-DT model, respectively.

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Fig.6. Confusion Matrices Results for the N-SNN and W-SNN using Small and Large datasets.

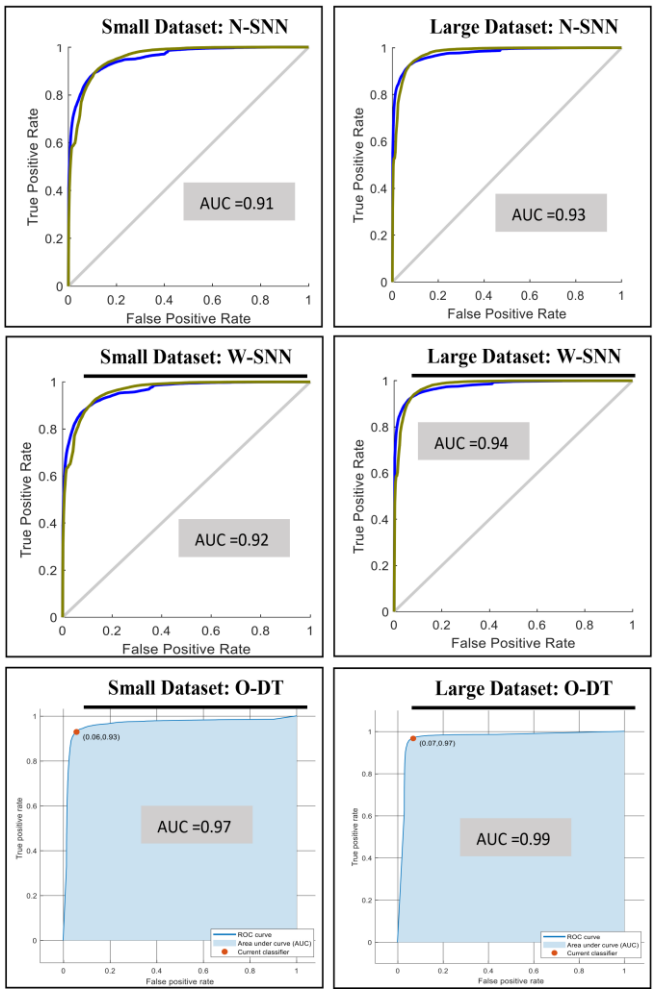

Fig.7. Analyzing ROC Results for the N-SNN and W-SNN using Small and Large datasets.

V. Conclusions

In this work, we presented a system for phishing websites detection using URL addresses. Our system solves this problem as a binary classification problem where websites are classified into either authentic or phishing websites. The system utilized machine learning techniques such as shallow neural networks and decision trees to learn data patterns in websites URLs. Full implementation details were provided including system model, testbed environment and configurations. The system has been evaluated on the Phishing Websites Dataset which includes small (balanced-class) and large (unbalanced-class) datasets. Our analysis showed that decision trees can provide 97.40% classification accuracy on the almost balanced-class dataset whereas it showed less accuracy (96.3%) on the unbalanced-dataset. Our future works will be directed towards improving the classification accuracy of our system. Other machine learning techniques will be used and evaluated to improve classification accuracy. We also aim at improving the computational efficiency of our system by providing an optimized implementation using fast machine learning frameworks built using low-level programming languages.

Finally, table 4 compare the performance of our model based gentle_boosted ensemble (O-DT) with other state of art models in terms of several reported metrics. The comparison reveals that our model is comparable and superior in detecting phishing website in an intelligent automated approach.

Table IV. Comparison with other state of art models.

Phishing Detection Model	Accuracy %	F1-Score %	AUC %
Hamza et. al, [29]/2021	95.70	-	-
Peng et. al, [30]/2018	86.61	92.96	-
Saha et. al, [31]/2020	93.00	-	-
Patil et. al, [32]/2018	96.58	-	-
Barlow et. al, [33]/2020	94.16	91.30	-
Patil et. al, [32]/2018	95.70	92.30	-
Proposed Model	97.40	96.30	99.00

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